

12

$$\begin{aligned}
 w_{\text{by}} &= m(\phi_f - \phi_i) \\
 &= 1 \left(-\frac{5GM}{2R} - \left(-\frac{GM}{R} \right) \right) \\
 &= -\frac{3GM}{2R}
 \end{aligned}$$

13 $\Delta T = 2 \text{ K min}^{-1}$